

ClauseGuardRAN: Citation-Grounded Multimodal RCA for O-RAN Troubleshooting

Project plan and weekly timeline

Sep 1 – Dec 11, 2026 · status Oct 1, 2026

1. Project Overview

- Detect faults from KPI time series and from alarms, logs, or natural-language context.
- Retrieve supporting passages from a curated O-RAN / 3GPP specification corpus (RAG).
- Produce a root-cause explanation with citations to KPI evidence, text evidence, or spec chunks.

Data: TelecomTS (KPI windows, NL descriptions, fault and RCA labels); O-RAN / 3GPP specs with manifest, versions, and clause-level chunks.

Experiments: Task 3 — KPI-only and text-only baselines (no RAG). Task 4 — sequential multimodal RCA with a four-row ablation. Task 5 — KPI-text fusion model vs the Task 4 sequential pipeline.

Outputs are suggestions for a human reviewer. The system does not act on a live network.

Pipeline

```
[KPI window] → time-series model → fault_class, top_KPI_deltas
[Alarms / logs / NL] → summarize
→ RAG query from fault, counters, alarm text
→ retrieve spec chunks → LLM / LoRA-SLM → RCA + citations
evidence: kpi_stats, alarm_ids, spec_chunk_ids
```

Progress (Oct 1, 2026)

- Weeks 1–3: corpus, chunking, Qdrant index (oran_specs_nemotron_vl_2048, 10,715 chunks), /health, /search, grounded /query.
- Week 4 TelecomTS ingestion slipped; catch-up in Week 5.
- /rca and /evaluate still return 501.

Task numbering in this document: fusion model is Task 5 (was Backup Task 1). FastAPI / MCP / demo packaging is Backup Task 1 (was Task 5).

2. Main Tasks

Task 1: O-RAN / 3GPP knowledge base and retrieval

Curate O-RAN and 3GPP documents with a versioned manifest. Parse PDFs, chunk on clause boundaries, embed, index in Qdrant, and expose /search and grounded /query with citations. Same retrieval core for all later tasks.

Task 2: Dataset and golden evaluation set

Ingest TelecomTS into `scenarios.jsonl` (`scenario_id`, `kpi_window`, `nl_context`, `fault_label`, `rca_label`, etc.) with a leakage-safe train/test split. Build a golden set of hand-written `spec_qa` and `rca` rows (≥ 30). Run Recall@k / MRR and RCA metrics via a repeatable script that writes artifacts.

Task 3: KPI and text fault-detection baselines

KPI path: $X_{\text{KPI}} \rightarrow \text{time-series model} \rightarrow \text{fault / root cause}$. Text path: `logs / alarms / NL` $\rightarrow$ LLM $\rightarrow$ fault / root cause. Report fault-class and RCA-class accuracy on a held-out split (no specification retrieval in these baselines).

Task 4: Citation-grounded multimodal RCA and SLM comparison

Sequential RCA stage:

Fault prediction + KPI evidence + text evidence + retrieved spec chunks $\rightarrow$ LLM $\rightarrow$ root-cause explanation

Each answer includes the predicted fault and evidence (KPI stats, alarms/logs, or cited spec excerpts).

Four configurations on the Task 2 golden set (all use the same frozen index, retriever, and held-out split; all include RAG for RCA generation):

Configuration	What it tests
KPI + text + RAG + base LLM	Reference sequential RCA: both modalities in the prompt plus retrieved spec chunks.
KPI + text + RAG + LoRA SLM (same retrieval and prompt template)	Whether a domain-adapted small model matches the base LLM when evidence is identical (only the checkpoint changes).
Text + RAG (no KPI summary in the RCA prompt)	Whether KPI statistics improve RCA labels and citations beyond alarms/NL text alone.
KPI summary + RAG (minimal alarm/NL text)	Whether textual context improves results beyond compact KPI evidence.

Unimodal fault detection without RAG stays in Task 3. Retrieval quality (Recall@k, MRR) is scored on `spec_qa` golden rows; RCA accuracy, citation coverage, and citation faithfulness on `rca` rows; latency and memory on all runs. Error analysis: retrieval miss vs generation miss vs citation mismatch.

Task 5: Multimodal KPI-text fusion model

$F_{\text{KPI}} \otimes F_{\text{Text}} \rightarrow \text{fusion} \rightarrow \text{fault / root cause}$

Compare fusion against the Task 4 sequential pipeline (reference: KPI + text + RAG + base LLM) on the same held-out split.

3. Backup tasks

Backup Task 1: Package service (FastAPI + MCP + demo)

Wire modules behind one service with logging and latency visibility. FastAPI: /rca, /query, /evaluate, health. MCP: thin tools over the same core (e.g. search_o_ran_specs, run_rca, summarize_kpi_window). Demo from KPI scenario to cited RCA; report with limitations.

Backup Task 2: Multi-hop retrieval

Follow cross-spec references (e.g. A1 → E2AP) via a chunk graph; measure Recall@k on compound questions vs single-hop.

Backup Task 3: Cross-dataset generalization

Test on SpotLight Open RAN KPIs or TeleLogs RCA; check transfer beyond TelecomTS.

Backup Task 4: Dry-run A1 policy assist

Schema-validated policy candidate from a scenario; human approval only; no live RIC submission.

4. Weekly timeline

Weeks run Tuesday–Monday from Sep 1, 2026. Thanksgiving week off. Submit Friday, Dec 11, 2026.

Week	Dates (2026)	Task	Goal	Status
1	Sep 1 – Sep 7	Task 1	Corpus manifest and PDF parser	Delivered
2	Sep 8 – Sep 14	Task 1	Chunking, embeddings, Qdrant, /search	Delivered
3	Sep 15 – Sep 21	Task 1	Grounded /query; corpus freeze	Delivered
4	Sep 22 – Sep 28	Task 2	TelecomTS ingestion	Slipped
5	Sep 29 – Oct 5	Task 2	TelecomTS → scenarios.jsonl, leakage-safe split, manifest	Current
6	Oct 6 – Oct 12	Task 2	Golden set ≥30 rows; Recall@k / MRR runner	Planned
7	Oct 13 – Oct 19	Task 3	KPI and text baselines; GPU smoke test	Planned
8	Oct 20 – Oct 26	Task 4	Sequential RCA on ≥5 scenarios; LoRA JSONL draft	Planned
9	Oct 27 – Nov 2	Task 4	One QLoRA run (Llama 3.1 8B); fixed retrieval	Planned
10	Nov 3 – Nov 9	Task 4	Four-config ablation; freeze Task 4	Planned
11	Nov 10 – Nov 16	Task 5	Train fusion model on Task 2 split	Planned
12	Nov 17 – Nov 23	Task 5	Fusion vs Task 4 sequential; write-up	Planned
—	Nov 24 – Nov 30	—	Thanksgiving	Off
13	Dec 1 – Dec 7	Report	Figures and report draft; Backup Task 1 demo if time	Planned
14	Dec 8 – Dec 11	Report	Rehearse; submit Dec 11	Deadline